

YAVUZ BAHADIR INCI

Mechanical Engineering Student

(+90) 5519772727

y.inci2022@gtu.edu.tr

<https://www.ybinci.com/>



ABOUT

Mechanical Engineering student at Gebze Technical University ,My background in rowing and cycling sparked a deep interest in aerodynamics and fluid mechanics, driving my passion for aerospace design and innovation

EDUCATION

SEPT 2022-Ongoing

BSc Gebze Technical University, Mechanical Engineering

MAINTENANCE INTERN | ÇOLAKOĞLU METALURJİ A.Ş.

JULY2026-AUGUST2026

Supported maintenance operations for industrial mechanical equipment, including preventive maintenance, fault diagnosis, and equipment reliability improvement.

Observed maintenance practices for pumps, conveyors, gearboxes, motors, and hydraulic/pneumatic systems in a heavy industry environment.

MECHANICAL DESIGN ENGINEER | DGA DRONE

JULY2025-JANUARY2026

Led and contributed to the design, manufacturing, and development of advanced UAV projects, including Hybrid **Firefighting**, **Agricultural Spraying**, **Facade Cleaning**, and **Defense** Drones. Participated in mechanical design, CAD modeling, structural analysis, and prototype manufacturing using SolidWorks and ANSYS.

Contributed to **R&D** by developing a **glass-breaking mechanism**, **optimizing spray nozzle geometry**, and designing a **modular robotic arm** for industrial and defense applications.

FREELANCE MECHANICAL & SIMULATION ENGINEER

2023-Ongoing

- Delivered CFD, FEA and mechanical design services using SolidWorks, Fusion 360 and ANSYS Structural.
- Completed multidisciplinary simulation projects for engineering students and professionals.
- Developed aerodynamic CFD models for UAV platforms and atmospheric flow analyses.
- Executed architectural wind distribution studies for site planning using CFD.
- Performed structural FEA for mechanical and architectural design tasks.

Designed robotic arms, excavator mechanisms and marine vessel components in Fusion 360.

TECHNICIAN | İNCİ ELEKTRİK

2016-2022

Developed strong hands-on expertise in power tools and workshop operations, gaining a technician-oriented engineering perspective through direct involvement in installation, assembly, and maintenance tasks alongside professional electricians.

PROJECTS

CONTROL SYSTEM DESIGN PROJECT | Dual-Rotor T-Beam Balancing System

- Designed and developed a dual-rotor balancing system for attitude stabilization using Arduino-based control.
- Developed the mathematical model and state-space representation of the system.
- Performed controller design and stability analysis using MATLAB/Simulink.
- Integrated sensors, ESCs, and BLDC motors to validate the control system on a physical prototype.
- Contributed to mechanical design, experimental validation, and technical documentation.

MECHANICAL DESIGN PROJECT FEMOR Rehabilitation System - Designed a modular lower-limb rehabilitation system with adjustable anthropometric dimensions. - Performed FEA validation for 70–250 kg load cases using ANSYS Workbench. - Optimized the mechanical design using telescopic linkages. - Prepared and presented the complete technical report and FEA analysis.		
TECHNICAL SKILLS - FUSION 360 - SOLIDWORKS - ANSYS - MATLAB		
LANGUAGE - TURKISH NATIVE - ENGLISH B2		
ACTIVITIES GTU Rowing Crew, Team Captain SEPT 2023-Ongoing Led the university's rowing team in training and national and international competitions. Organized club events and served as president of the Rowing Club. GETAK Search and Rescue, Volunteer Member DEC 2023-2024 Participated in search and rescue training programs and emergency preparedness operations. Collaborated with team members during simulation exercises and university-level rescue drills. Galatasaray Rowing Team, Former Athlete (2 years) Competed in national and international rowing championships, representing Galatasaray. Sakarya BBPRO Team, Former Athlete (+3 years) Competed in national and international cycling races, representing Sakarya BB Pro Team.		
ORGANIZATION GTU Sports Summit, Organizer Coordinated and managed the GTU Sports Summit, a multidisciplinary event bringing together student-athletes, coaches, and professionals to discuss the intersection of sports, science, and leadership. Oversaw event planning, speaker invitations, and social media outreach		MAY 2025
HOBBIES I actively create educational video content on Instagram and YouTube, focusing on aircraft, aerospace systems, engineering concepts, coaching insights, and popular science topics. I take pride in making complex ideas more accessible to a broader audience. My background as a former professional athlete in rowing (Galatasaray Rowing Team) and cycling (Sakarya BB Pro Team) sparked a strong passion for aerodynamics and flight mechanics, shaping my long-term interest in aerospace engineering. In addition, I am deeply interested in scientific research and regularly analyze new studies to stay informed about the latest advancements in science and technology.		